# DESIGN AND ANALYSIS OF ALGORITHM (CS-327)

# LECTURE – 7

## LINEAR SEARCH ALGORITHM

**Linear Search** is defined as a sequential search algorithm that starts at one end and goes through each element of a list until the desired element is found, otherwise the search continues till the end of the data set. Linear Search is also known as sequential Search. The complexity of algorithm is **O(N)**.

Example :-

**Size of array (N)**
**Key Element = X**

```
if( arr[i]= = x)
{
return x
}
```

| i | arr |
|---|---|
| 1 | 5 |
| 2 | 10 |
| 3 | 15 |
| 4 | 20 |
| . | 25 |
| . | 30 |
| N | N |

Value :-

| 5 | 10 | 15 | 20 | 30 | 35 | - | - | - | - | - | - | - | **N** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

Location:- 0 1 2 . . . . . . . . . . N

We have 2 inputs (array, element)

| LINEAR SEARCH ALGORITHM | TIME COMPLEXITY |
|---|---|
| int linearsearch(int arr[ ],int x) | |
| { | |
| for(int i=0; i <arr.length; i++) | 1 + (N+1)+N |
| { | |
| if(arr [i] = = x) | N |
| { | |
| return i; | (Never executes in worst case) |
| } | |
| } | |
| // x is not found | |
| return -1; | 1 |
| } | |
| | T(N) = 1 + (N+1)+N+N+1<br>= 3N+3<br>Considering Big O notation we will consider the worst case or term that has higher growth.<br>T(N) = O(N). |